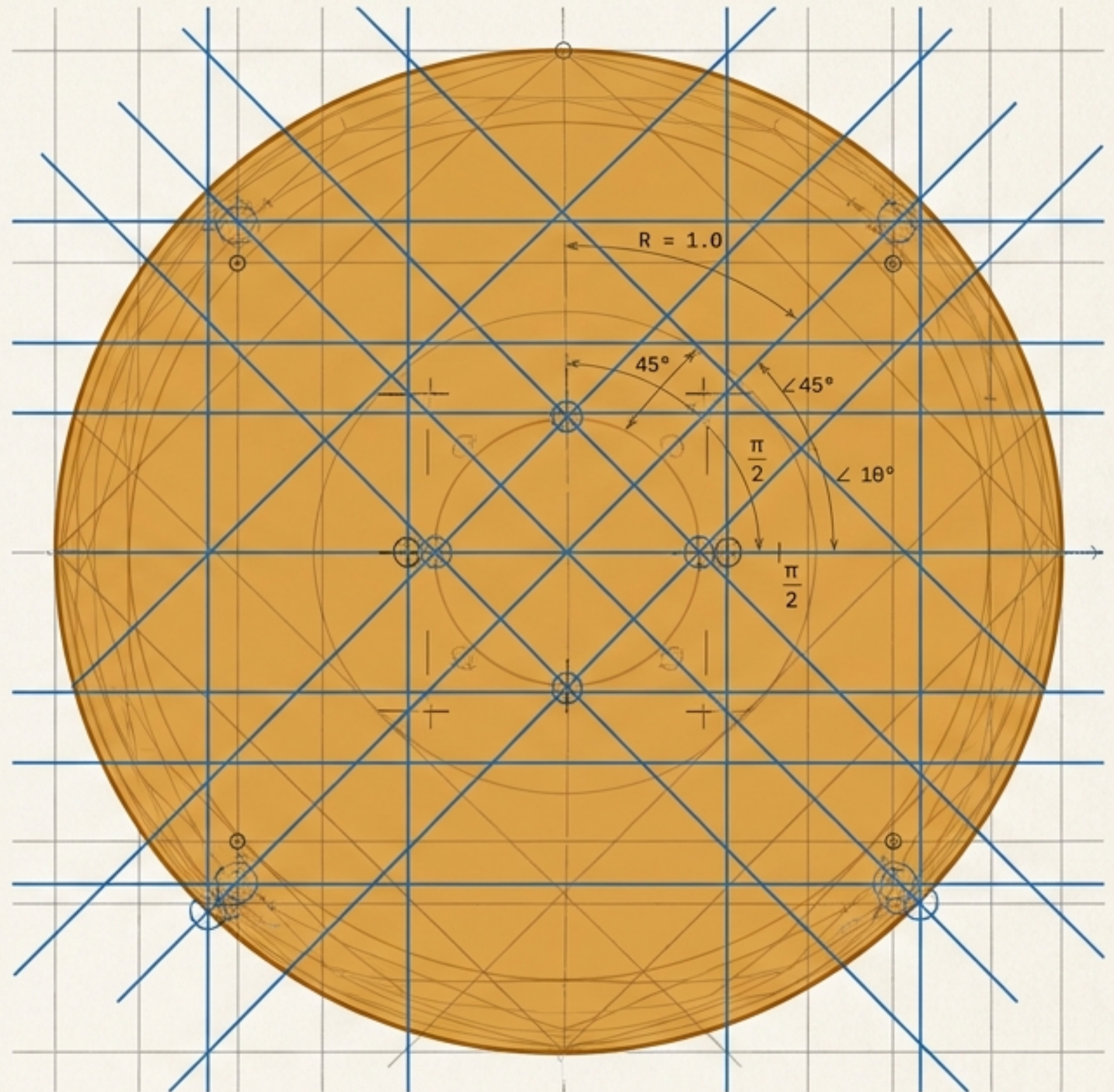


Rotation Before Number

Admissibility, Geometry,
and the Quotient Space of
Quantum Mechanics

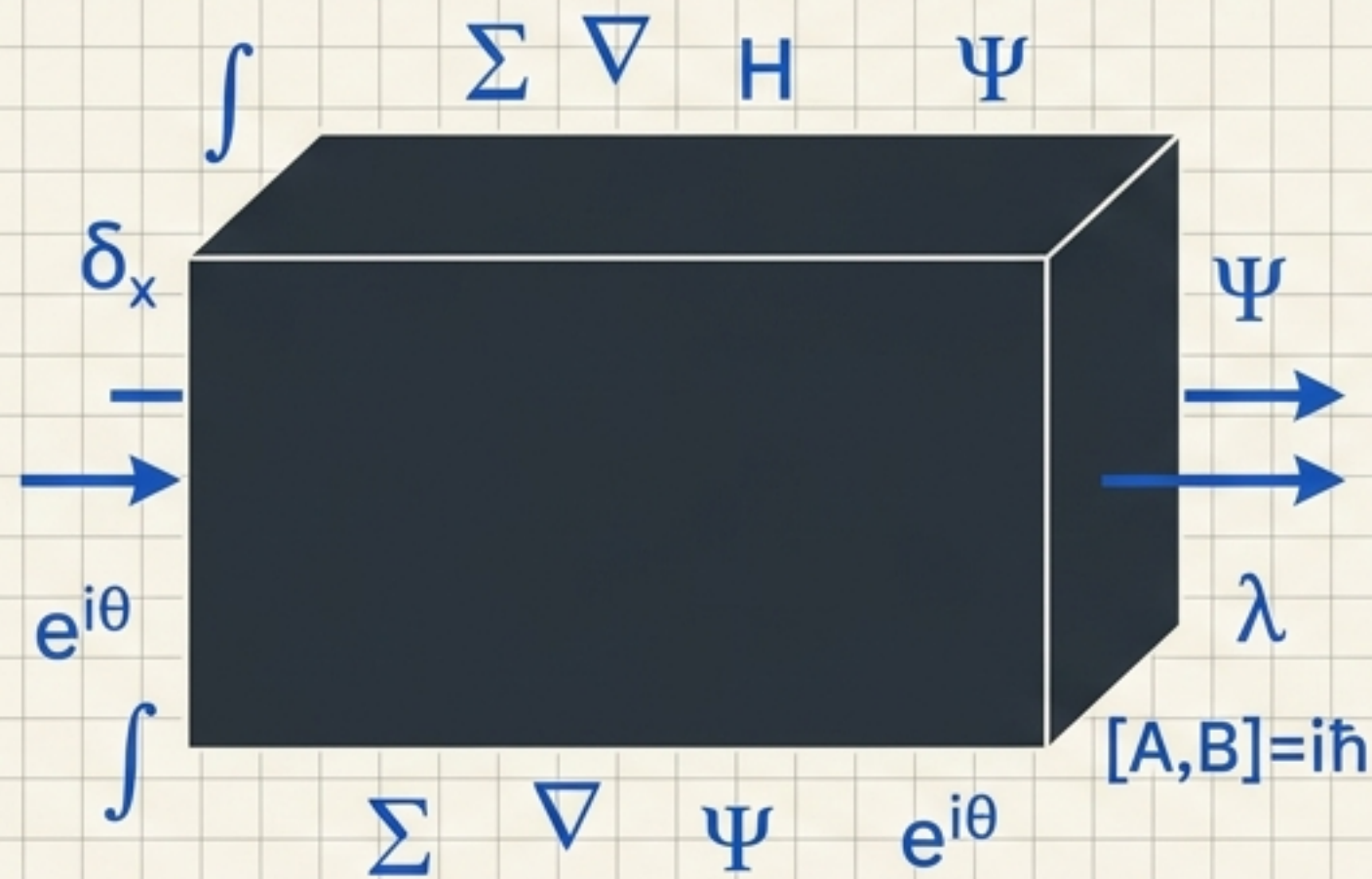
Why complex numbers are just
compressed geometry—and what that
reveals about physical reality.



Explanation Before Correctness

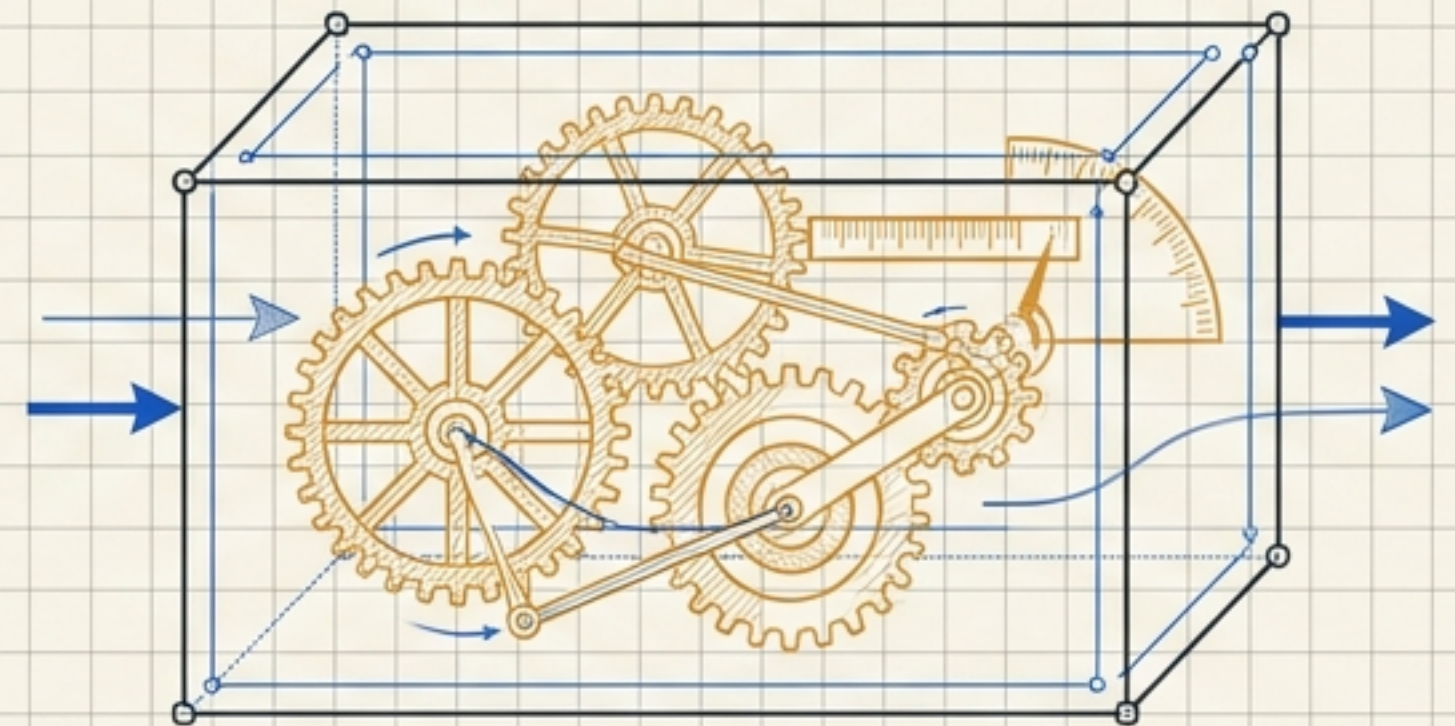
A formalism can yield perfect predictions long after the geometric intuition that produced it has been compressed into opaque notation.

Correct Representation



Produces right answers, hides the “why.”

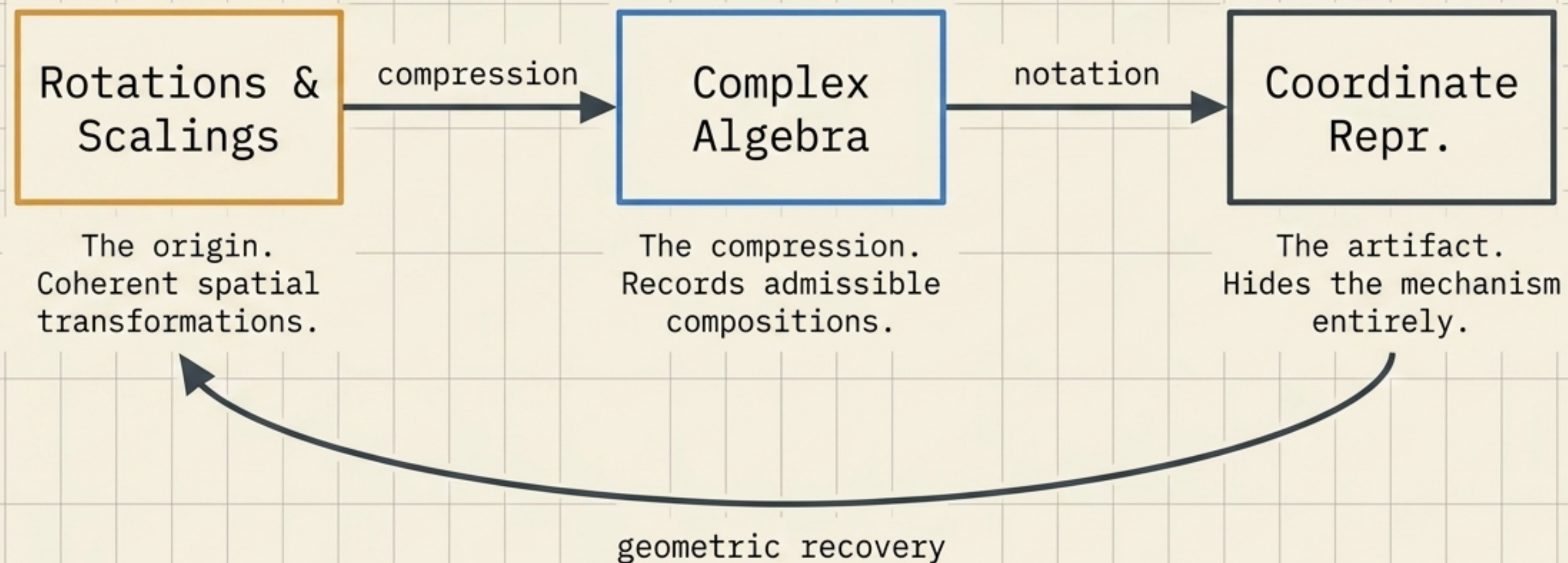
Explanatory Representation



Makes the generative mechanism visible.

Insight: The standard algebra of quantum mechanics is correct, but epistemically opaque. We must recover the generative mechanism.

Needham's Inversion: Geometry Generates Algebra



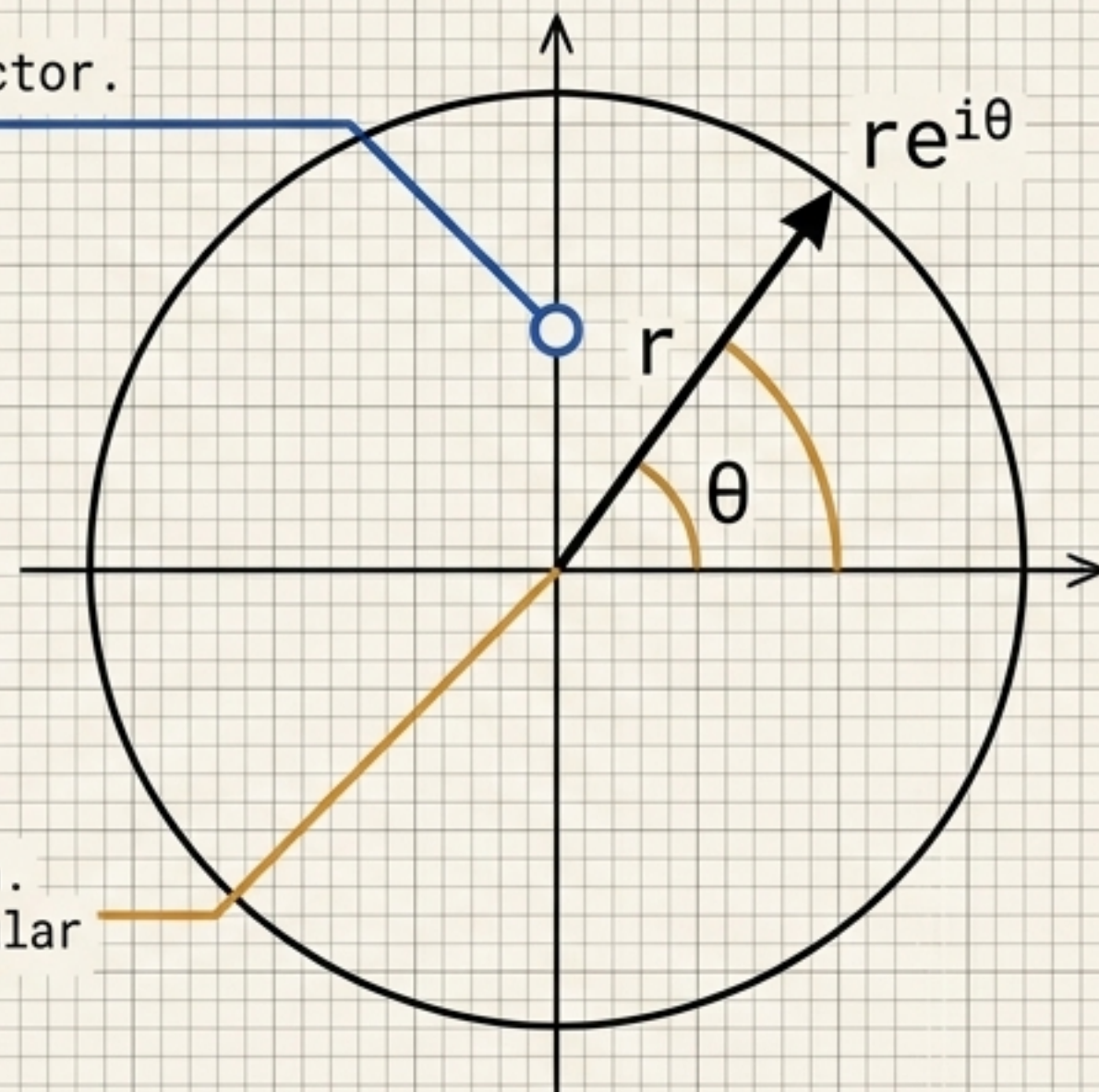
Takeaway: Complex algebra did not emerge from formal square roots of -1 . It emerged to record spatial rotations.

Decompressing the "Amplitwist"

$$z = r e^{i\theta}$$

Amplitude: The scale factor.
An independent radial transformation.

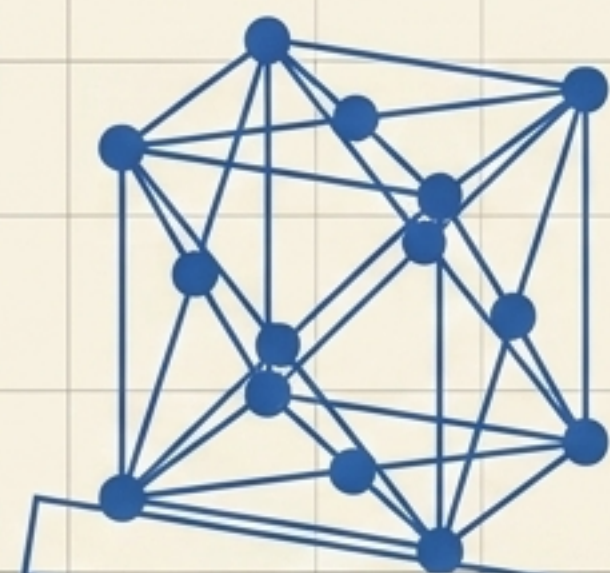
Twist: The rotation.
An independent angular transformation.



The Needham Principle: Transformations are ontologically prior to the coordinates used to describe them. The number merely records the process.

The Quantum Dimension Mismatch

Can we write Quantum Mechanics entirely with real numbers to expose the geometry?



Complex Space:
 $\mathbb{C}^2 \otimes \mathbb{C}^2$

Requires 8 real parameters.

Naive
Real Space:
 $\mathbb{R}^4 \otimes \mathbb{R}^4$



Requires 16 real parameters.
Naive extensions overcount.

The Obstacle: This dimensional excess represents mathematical distinctions that have zero measurable physical reality.

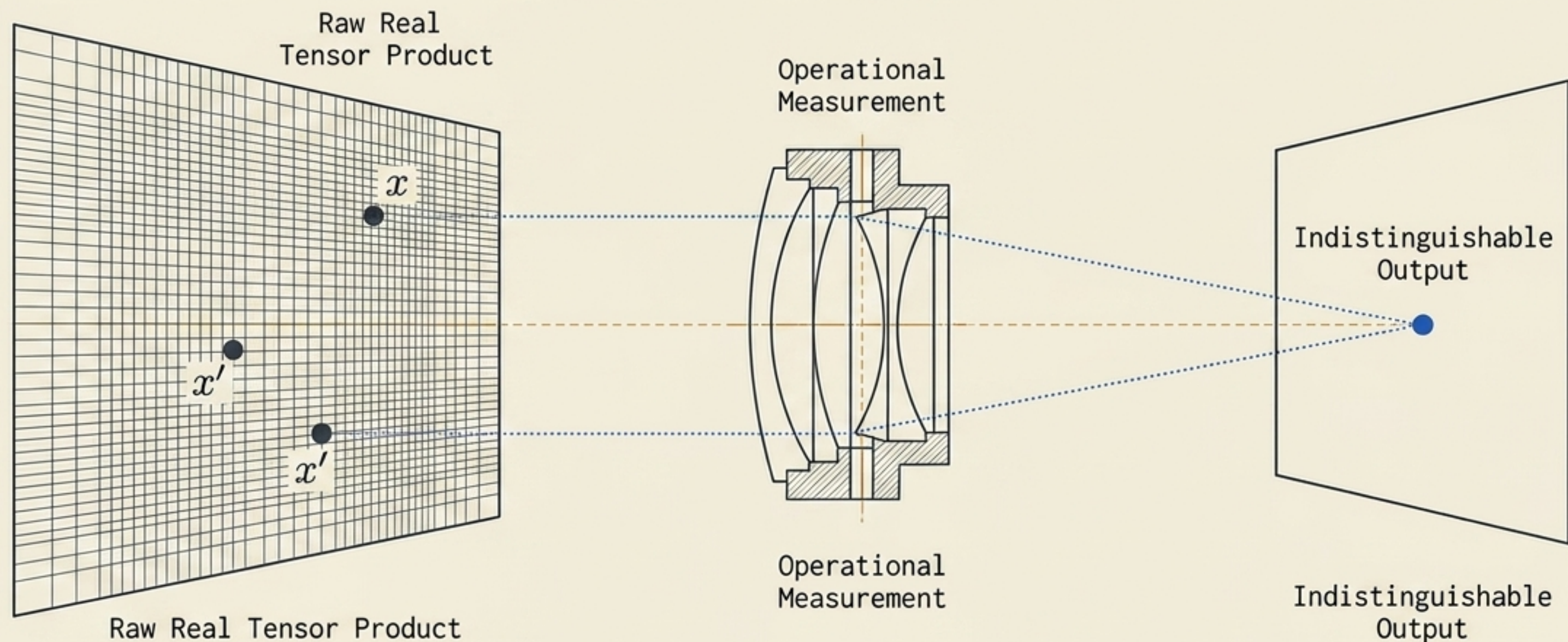
The Flag Space: Rotation Made Manifest

Mechanism: The Barrios et al. construction introduces a 2D "flag" system to track real and imaginary parts, recovering the geometry.

	Complex QM	Real Flag Space QM
1	Imaginary Unit: i	Quarter-Turn Matrix: $J_F \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ 
2	Multiplication: $(a + ib)(c + id)$	Matrix Algebra: $(aI_F + bJ_F)(cI_F + dJ_F)$
3	Global Phase: $e^{i\alpha}$	Flag Rotation: $R_F(\alpha)$ 

Conclusion: 'i' was never just a number. It was always a geometric quarter-turn. The complex algebra reappears purely as real matrix operations.

The Admissibility Gap in Composite Systems



The Problem: In composite systems, the real tensor product creates states that look mathematically distinct on paper, but are perfectly indistinguishable in the lab.

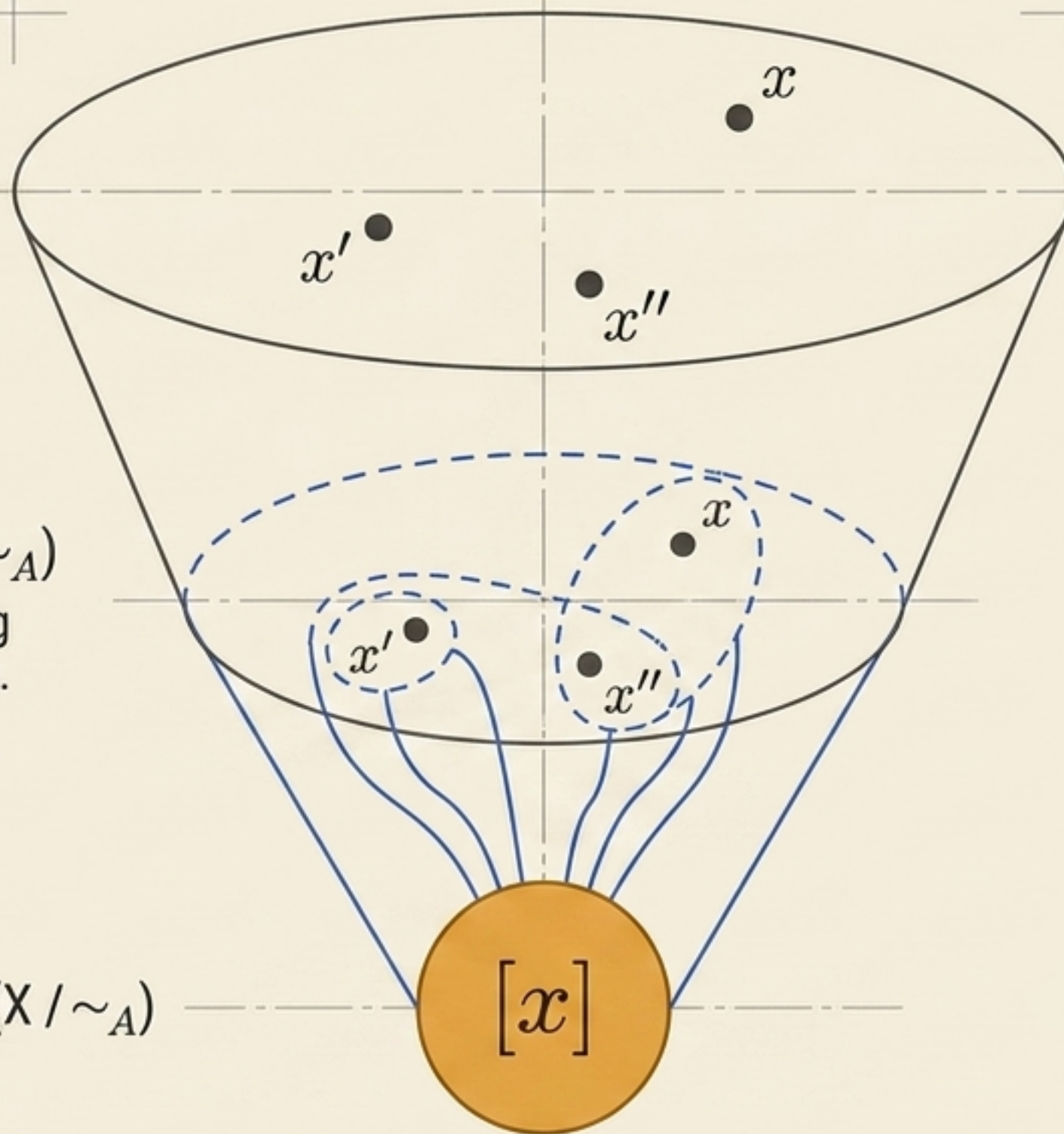
Definition: Admissibility: A rule determining which mathematical distinctions correspond to physically realizable differences, and which are merely representational artifacts.

The Quotient as Repair

Raw Real Tensor Product

Admissibility Equivalence ($\sim_A$)
Deliberately identifying
operationally identical states.

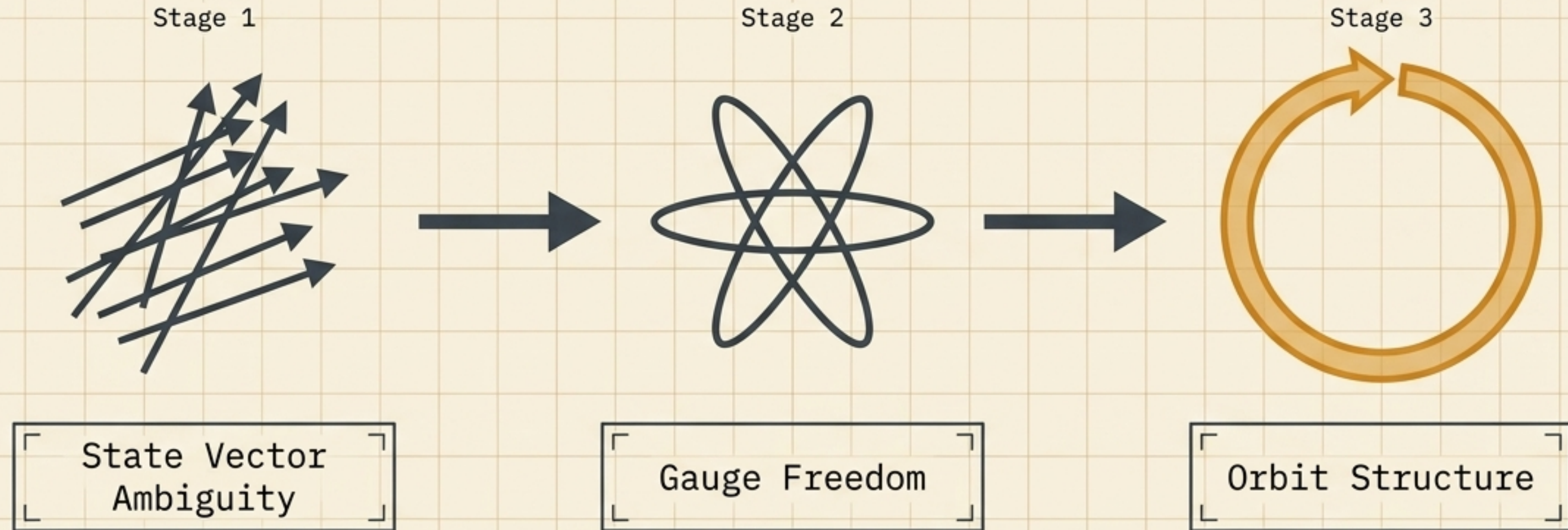
Repaired Composite Space ($X / \sim_A$)



Insight: This is not correcting an error. It is removing representational distinctions that were never genuine to begin with.

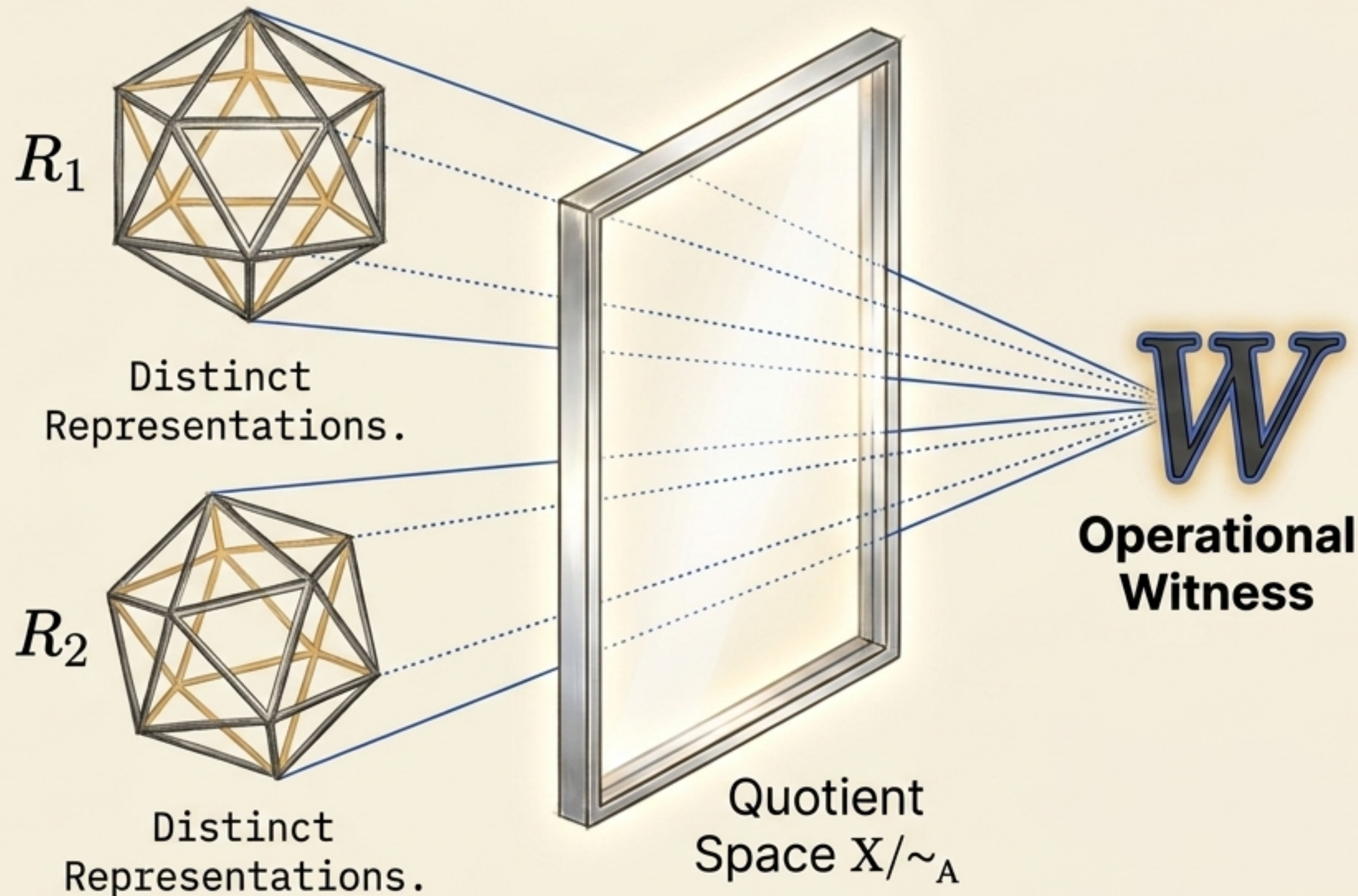
The Conservation of Ambiguity

When we quotient the state space, ambiguity does not vanish. It relocates.



The Physical Rule: Physical State = Gauge Orbit
(Not Representative Vector).

Witnesses: The Filter of Reality



Definition:

A witness (W) is any operationally accessible quantity (a probability, an expectation value).

Witness Invariance:

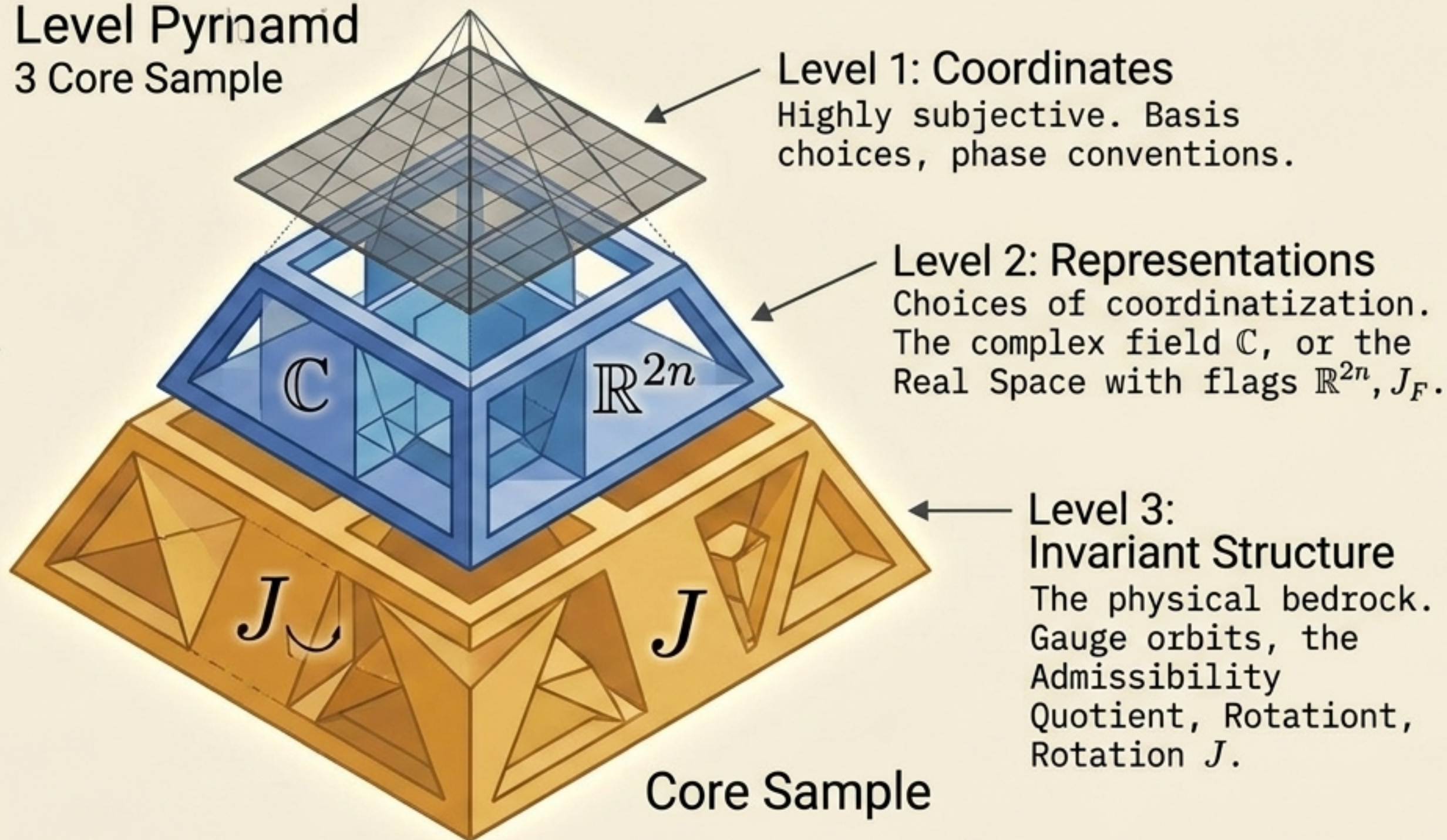
If $W(R_1) = W(R_2)$ for every admissible witness, then R_1 and R_2 are physically identical.

The Slogan:

What persists across representations is what is real. The coordinate representation is the artifact.

The Ontological Hierarchy

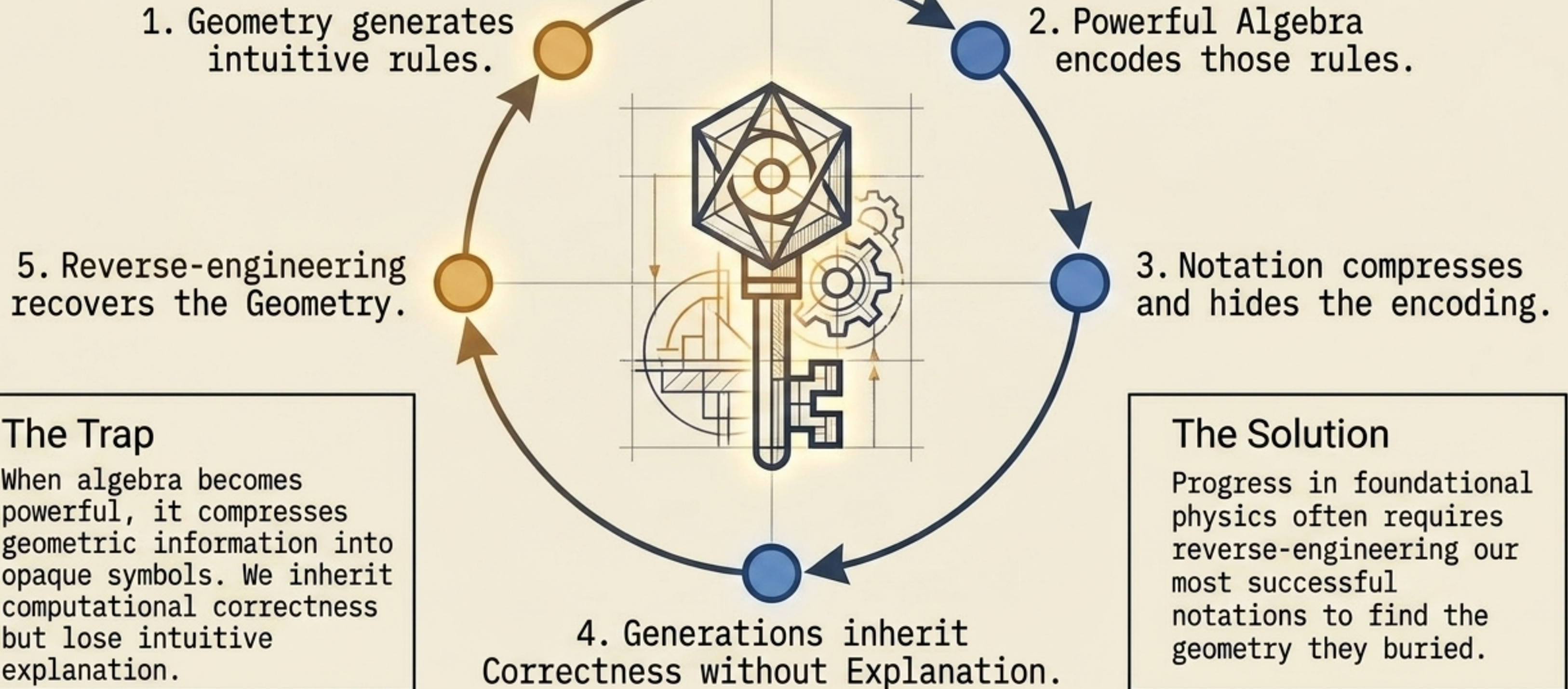
Level Pyramid
3 Core Sample



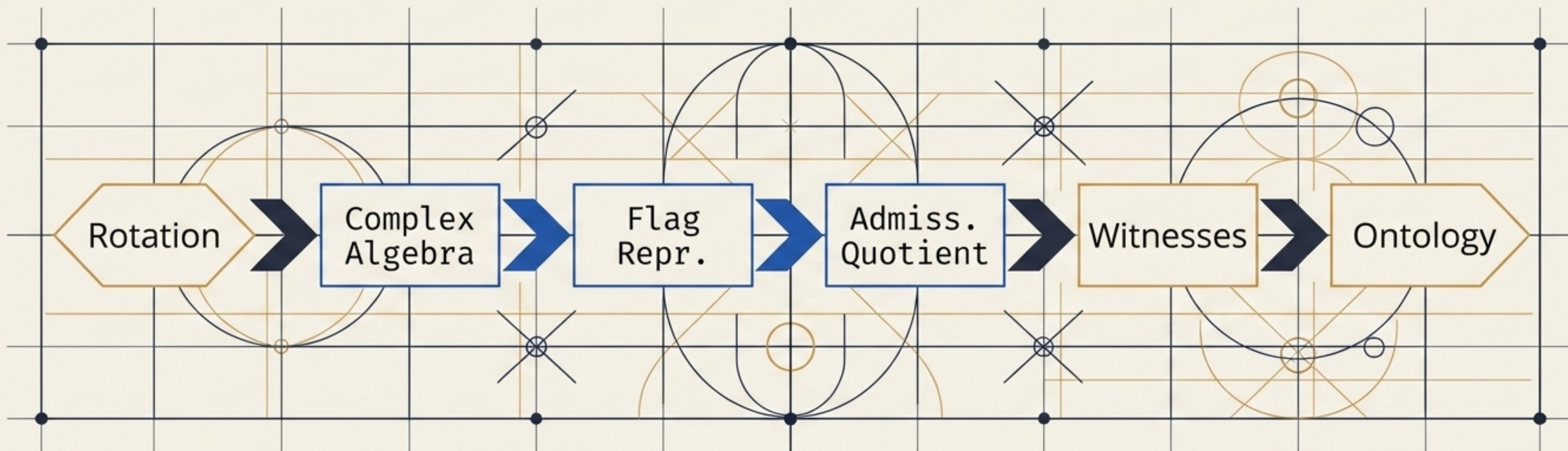
Physical ontology belongs to the deepest layer alone.

Complex and real formulations are not different theories; they are just two coordinate systems draped over the same invariant reality.

The Cycle of Abstraction



Reality is the Invariant



Block 1 (Summary)

Complex numbers are incredibly useful, but they are not strictly necessary to describe quantum mechanics. The rotational geometry they encode, however, is fundamental.

Block 2 (Final Thought)

The geometry was always there. The quotient merely reveals which distinctions belong to reality, and which belong only to our descriptions.